ChatScale — Scope & Honesty Notes

Faculty handout · excerpted from docs/REPORT.md §7 and §9 · full report: docs/REPORT.md

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What this project deliberately does not claim

Three rows of the original thesis table (PLAN.md §1) needed real, separate physical machines — Cluster Autoscaler adding whole nodes, an actual availability-zone failure, a managed database failover — and a fourth needed a real cloud invoice to measure economics rather than estimate them.

Why these are missing, honestly: Azure for Students verification was attempted and never resolved (SheerID lost the original request, a resubmission ran into further account/product-page friction; full account in PLAN.md's r4 amendment and docs/CHANGE_LOG.md CHG-0009/CHG-0033). Rather than restart the same signup/verification process with a second cloud provider (DigitalOcean or GCP, both considered — PLAN.md's own risk R-12 named both as fallbacks), the decision was to finalize this project's scope at local Kubernetes. This is a stated, reasoned engineering tradeoff — not an oversight, and not something quietly worked around by claiming local results as if they answered a cloud-scale question they don't.

What this means concretely:

- **No Cluster Autoscaler comparison.** The HPA comparison (§6) is pod-level only (more pods on existing nodes) — real elasticity (nodes appearing to host pods that don't fit anywhere yet) needs a cloud provisioning node pools on demand.
- **No zone-failure or managed-database-failover chaos test.** The pod-kill test (§2.3) is the one chaos scenario that's fully meaningful locally; “kill a zone” has no local equivalent when every “node” is a container on one machine.
- **No real cloud invoice.** The cost model (§8) is an estimate from measured resource requests and published pricing, explicitly labelled as such.

Methodology notes and honesty

Consistent with how this project has recorded its work throughout (see docs/CHANGE_LOG.md's entries, which keep debugging stories rather than cleaning them into tidy-looking successes):

- Every number in the report traces to a docs/evidence/*.md file with the exact command used to reproduce it. Nothing is estimated unless explicitly labeled as an estimate (§8).
- Bugs found while measuring (the HPA double-division bug, §6; the registration-burst CPU spike mistaken for chat load, §6) are reported as part of the finding, not edited out.
- Sample sizes and simplifications (n=1 vs PLAN.md's n=3 in §5) are stated, not hidden.
- Three results are explicitly *not* claimed (§7 above), with the reason why.